MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR, MARK 121 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 121 Mod 0 Rocket Motor described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Mark 121 Mod 0 Rocket Motor (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

SPECIFICATIONS

Naval Air Systems Command (Code Ident 30003)

AS 3885	Propellant, N-47 (U)

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

673AS300	Rocket Motor, Mark 121 Mod 0

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Mark 121 Mod 0 Rocket Motor shall conform to Drawing 673AS300.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the requirements of table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Propellant N-47 (AS 3885).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0 % Rel. Humidity		22 % Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	149	-	149	-	149	-
E_m (elongation at S_m) %	77	21.3	-	10	-	18.5	-
Y_m (Young's modulus) psi	77	960	1800	960	2862	960	2880
B_r strand burn rate at 1000 psi) in/sec	77	0.69	0.84	0.69	0.89	0.69	0.89

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the requirements of AS 3885 when tested in accordance with MIL-A-85097.

3.3.2 Ballistic performance. The ballistic performance shall conform to table II.

TABLE II. Ballistic performance requirements.

Characteristic	Value		Temperature (± 5°F)
	Min	Max	
Rocket motor impulse (lb-s)	30.9	—	−40, 160
Action time (ms)	—	200	−40, 160
Peak thrust (lb)	158	472	−40, 160
Ignition delay (ms)	-	20	−40, 160
Initiation pressure (psig)	400	600	−40, 160
Burn time (ms)	72	—	−40, 160
Impulse angle (deg)	77	83	−40, 160

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: −40° ± 5°F
b. Hot: 160° ± 5°F

4.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 4 hours
b. Temperature cycling: Not less than 5 hours

4.1.3 Radiographic inspection. All components or assemblies shall be radiographed in one view.

5. PACKAGING

(See latest revision of MIL-A-85097.)

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6. NOTES

6.1 Intended use. The rocket motor is intended to insure proper trajectory divergence of the aircrew during an escape for the TAV-8B Aircraft Escape System.

6.2 Definitions.

6.2.1 Peak thrust. Peak thrust is the maximum value of thrust that occurs after the ignition peak thrust time interval.

6.2.2 Ignition peak thrust. Ignition peak thrust is the value of thrust that occurs within the first 0.030 seconds after first indication of thrust.

6.2.3 Impulse angle. Tangential angular relationship between the vertical and horizontal axes.

Preparing activity:
Navy - AS
(Project 1377-N911)

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